

Assignment No. 3. A complete pipeline that converts spoken language into a target language while preserving the speaker's voice characteristics and synchronizing lip movements in the corresponding video. Subsequently, audio and visual embeddings are extracted using ECAPA-TDNN and FaceNet, respectively, to train the baseline model.

Tasks

1. Create the pipeline - 10 marks
2. Extract embeddings from original and generated audio and visual data - 5 marks
3. Run the baseline and compare the results - 5 marks
4. Analysis to determine the amount of synthetic data required for each identity – 5 marks

(Bonus)

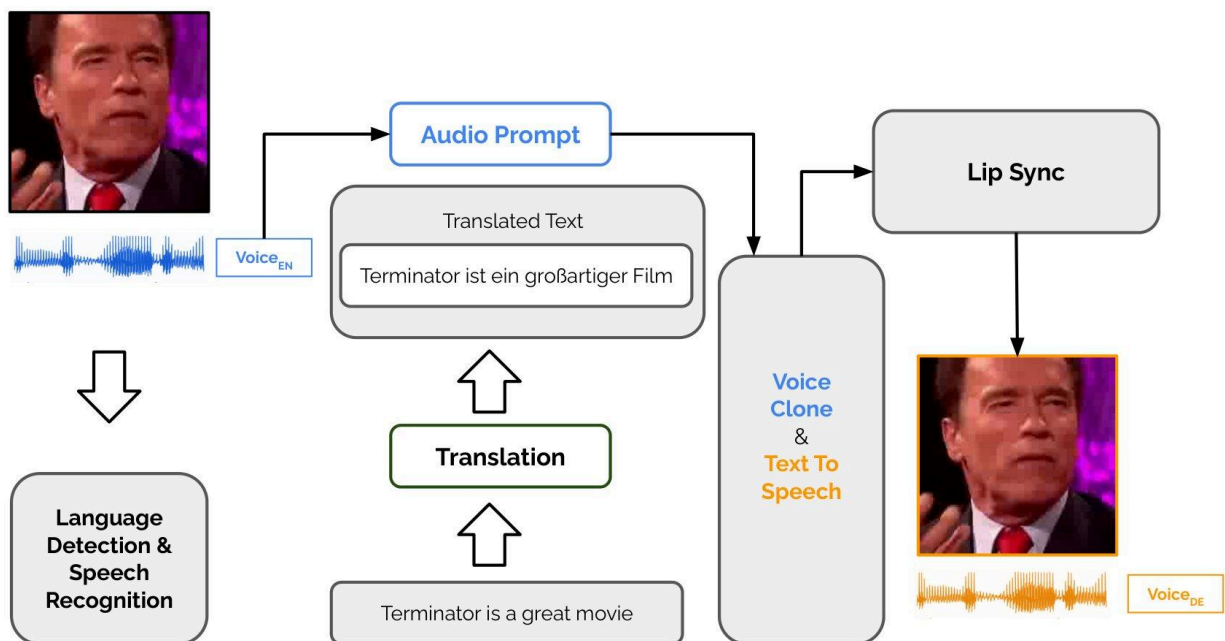


Figure 1. The pipeline takes a spoken video as input and first performs language detection and speech recognition to transcribe the audio. The transcribed text is then translated into a target language. Using the original voice as a prompt, a voice cloning and text-to-speech module generates speech in the target language while preserving the speaker's vocal characteristics. Finally, a lip synchronization module aligns the newly generated speech with the speaker's mouth movements in the video, producing a natural, translated output video.

Overview

1. Language detection and speech recognition. Language detection and speech recognition can be effectively performed using Whisper. It is important to consider that the model provides strong support for both English and German, enabling accurate transcription and language identification for these languages.

Translation. It can be achieved by sending the recognized text to a machine translation model, which converts it into the target language (e.g., English to German). Popular tools like Google Translate or DeepL can perform this task accurately.

Voice cloning and text to speech. Voice cloning and text-to-speech are used to generate speech in the target language while preserving the speaker's original voice characteristics (tone, pitch, style). Popular tools: Coqui TTS and Microsoft TTS. Details are as follow:

1. Voice Cloning

A short audio sample of the speaker is used to extract vocal features (e.g., pitch, timbre, accent).

This creates a speaker embedding that represents the person's voice.

2. Text-to-Speech

The translated text is then fed into a TTS model. The model generates speech in the target language while mimicking the original voice.

3. Output

The result should be natural-sounding audio that:

- Speaks the translated language (e.g., German)
- Sounds like the original speaker

Lip Sync. Lip sync is the final step where the generated speech (e.g., German audio) is aligned with the speaker's mouth movements in the video so it looks natural. Popular tools are Wav2Lip, LatentSync